

SlugShare

CSE 115A

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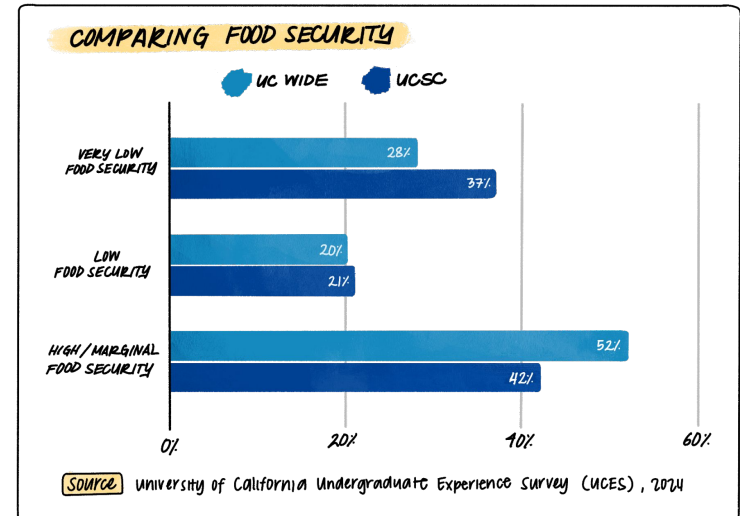
Product Owner: Hannah Dinh

Initial Scrum Master: Jordyn Salazar

SlugShare

Problem:

- 58% of students experience low food security at UCSC (10 pts higher than avg across UCs)
- Real world events that affect food accessibility (ex: ERCs & STARRS moving, Slugstop closing, undocumented students unable to get EBT, cost of living in Santa Cruz, government/EBT shutdown)
- There are students who end up with a lot of leftover dining points, which go back to the school.



An app connecting these students could be an additional option for people when in need of food and provide extra support during difficult times.

Project Scope

- High level goal(s) (for this project)
 1. Build a web app that connects UCSC students in need of food to those that are willing to share their extra points
 2. Be able to sign up as a user and have settings and requests be persistent across sessions
 3. Provide a dashboard where users can make and receive requests
 4. Have access to point spending tools so users can have a estimate of how much they can spend on others and themselves

Sprint 1

User stories:

- As a *user*, I want to sign up and log in so I can securely access the application (2)
- As a *user*, I want a dashboard where I can see my points balance, a button to view other people's requests, a button to make requests (1)(3)
- As a *user*, I want to view my profile information and edit my phone number, which can be shared with the donor/requester for communication purposes. (2)
- As a *requester*, I want to be able to cancel and delete my requests before they get accepted (3)
- As a requester, I want to see the status of my request (pending, accepted, timed out) (3)

Spikes

- UI is likely simple at this point, explore shadcn/ui components for building a better dashboard interface.
- Explore available tools for setting up text notifications

Infrastructure tasks:

- Setup Github repository
- Setup Kanban board (Github Project)
- Set up environment

Sprint 2

User stories:

- As a *requester*, I want to see food locations and their prices and hours of operation. (3)
- As a *requester*, I would like to indicate whether I'm okay with meeting in person and/or receiving a QR code (1)
- As a *donor*, I want to receive a notification when there is a new request to go to a dining hall/market and choose to accept or decline a request. (1)(3)
- As a *donor*, I would like to be able to generate a QR code linked to my Slug Points to share with *requester*. (3)
- As a *donor/requester*, I want a tool that lets me filter out existing/queued requests their chosen dining hall/market/restaurant location, the amount of points being requested

Spikes

- Explore prediction models/algorithms

Infrastructure tasks:

- Acquire or configure tools for notifications
- Test donor and requester interactions
- Make sure backend can be expanded upon with new features
- A secure backend for storing and sharing contact information

Sprint 3

User stories:

- As a *donor* I would like for the dashboard to have an accurate up to date showing of how many points my account has (3)
- As a *donor*, I want a tool that projects my points spending for the rest of the quarter and see how many points I have leftover to share. (4)
- As a *user*, I would like to see information about food pantries on campus (1)
- As a requester, I want to be able to see the GET QR code on my dashboard after having a request accepted
- As a *requestor*, I want to edit my already posted requests before it gets accepted (3)

Spikes

- Explore ways that users can safely get into contact with each other outside of the web application.

Infrastructure tasks:

- Database indexing in order to apply filters
- Find and document all bugs and parts of the web application that needs adjusting.
- Create a filter for if the requestor would be okay with meeting in person and/or using a QR code.
- Make login only for UCSC emails

Sprint 4

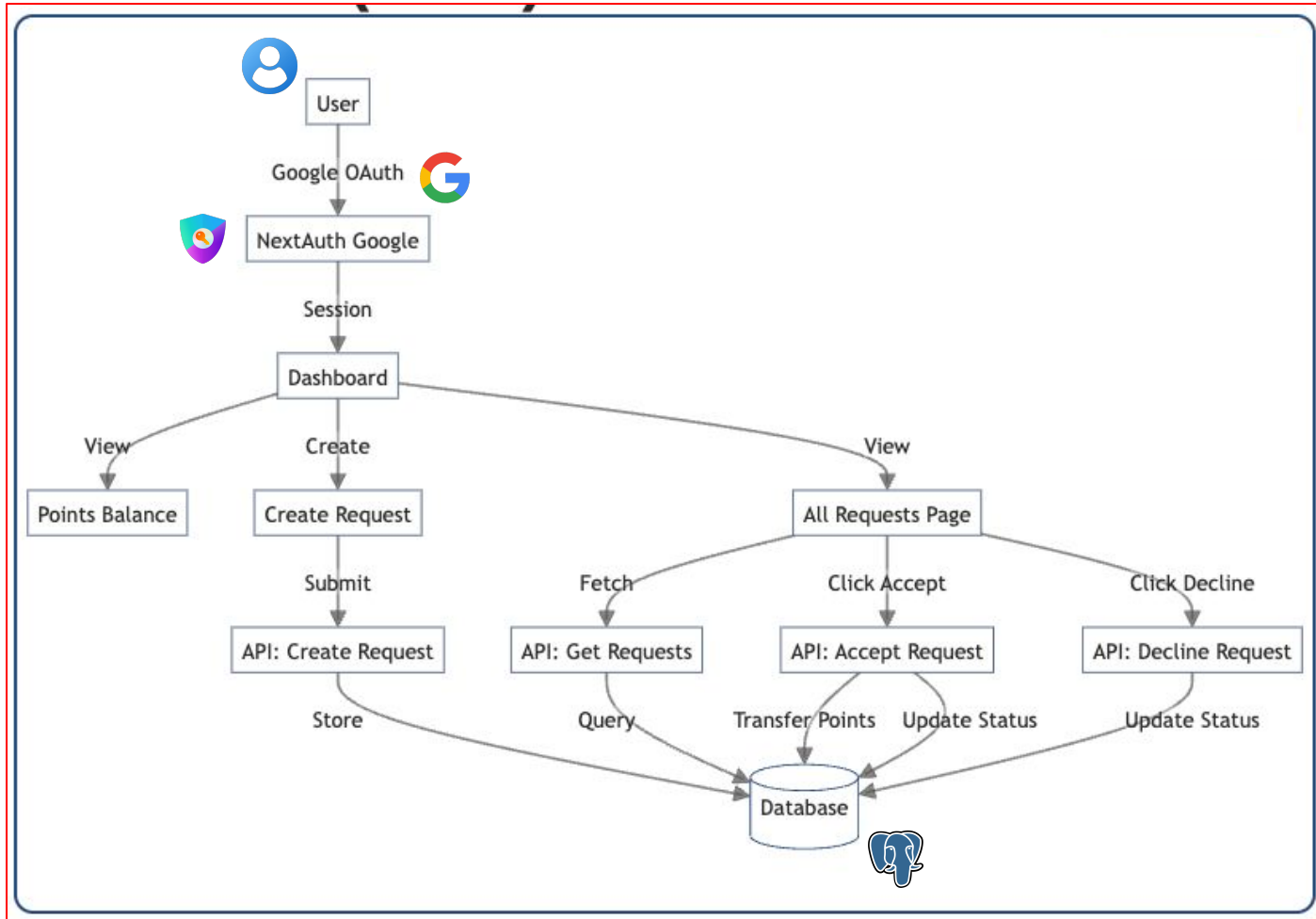
User stories:

- As a user, I want to be able to view a mobile specific version of the web application (5pts)
- As a donor, I would like my QR code to have a spending limit, time limit, and for the ability to be pulled (5 points)
- As a user I want push notifications and gmail notifications for requests accepts (5pts)
- As an admin, I want to view all requests to monitor system usage. (8pts)
- As a user, I want to be able to view a dark mode version of the web app (1 point)

Infrastructure tasks:

- An environment that is easy to test in.

Architecture



Technologies

Programming Languages: Typescript 

Web Framework: Next.js 



Development Environment: [Node.js](https://nodejs.org/), npm, Git and GitHub

Database: PostgreSQL , Vercel/Prisma 

UI: Tailwind CSS , React 

Authentication: NextAuth.js 

Challenges/Risks

Complex Feature set: Coordinating requests, accept/decline logic, and basic user communication

Unfamiliar Technology: Learning and integrating NextAuth.js, Prisma, and [Next.js](#), servers and databases.

Concurrency & Data Integrity: Maintaining consistent user sessions across logins and devices, prevent race conditions when multiple users interact with the same request.

Minimum Viable Product (MVP)

The minimum viable product for SlugShare is a functional web application that allows authenticated users to request dining points and be matched with donors who have sufficient points. Donors can view, accept, or decline requests, and all actions persist across sessions.

- As a *user*, I want to sign up and log in so that I can securely access the application.
- As a *requester*, I want to create a request so that donors can see my need.
- As a *donor*, I want to view requests and accept or decline them based on my available points.
- As a *user*, I want requests and balances to persist across sessions so that data is not lost when I log out.